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Problem Set 2.1

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Sir

(2) $\frac{d}{dx}y = \frac{x}{e^x}$

$\cosh(x) = \frac{e^x + e^{-x}}{2}$

$\sinh(x) = \frac{e^x - e^{-x}}{2}$

differentiating this equation with respect to x.

$\therefore \frac{dy}{dx} = \frac{d}{dx}\frac{x}{e^x} \quad g(x) \neq 0$

$= \frac{dy}{dx} = \frac{e^x \frac{d}{dx}x - x\frac{d}{dx}e^x}{(e^x)^2} = \frac{(1-x)e^x}{e^x e^x}$

(8) $f(x) = e^x \cdot \cos x$ differentiate this equation with respect to x.

$\therefore \frac{d}{dx}f(x) = f'(x) = e^x \cdot \frac{d}{dx}\cos x + \cos x \frac{d}{dx}e^x$

$= -\sin x \cdot e^x + e^x \cos x = e^x(\cos x - \sin x)$

(12) $g(\theta) = e^\theta(\tan\theta - \theta)$ differentiating this equation with respect to θ.

$\therefore \frac{d}{d\theta}g(\theta) = \frac{d\theta}{d\theta}\cdot(\tan\theta - \theta) + \tan\theta - \theta)\frac{d}{d\theta}e^\theta$

$= e^\theta(\sec^2\theta - 1) + (\tan\theta - \theta)e^\theta = e^\theta(\tan^2\theta + \tan\theta - \theta)$

(14) $f(t) = \frac{\cot t}{e^t}$ differentiating this equation with respect to t.

$\therefore \frac{d}{dt}f(t) = f'(t) = \frac{d}{dt}\frac{\cot t}{e^t} = \frac{e^t \frac{d}{dt}\cot t - \cot t \frac{d}{dt}e^t}{(e^t)^2}$

$$= \frac{-e^t \operatorname{cosec}^2 t - e^t \cot t}{(e^t)^2} = \frac{-e^t(\operatorname{cosec}^2 t + \cot t)}{e^t \cdot e^t}$$

$$= \frac{\operatorname{cosec}^2 t - \cot t}{e^t} = \frac{\frac{1}{\sin^2 t} - \frac{\cos t}{\sin t}}{e^t} \Rightarrow$$

(15) $J(v) = (v^3 - 2v)(v^{-4} + v^{-2})$ differentiating ....

$$= (v^3 - 2v)\frac{d}{dv}(v^{-4} + v^{-2}) + (v^{-4} + v^{-2})\frac{d}{dv}(v^3 - 2v)$$

$$= (v^3 - 2v).(-4v^{-5} - 2v^{-3}) + (v^{-4} + v^{-2}).(3v^2 - 2)$$

$$= -4v^{-2} - 2 + 8v^{-4} + 4v^{-2} + 3v^{-2} - 2v^{-4} + 3 - 2v^{-2}$$

$$= 6v^{-4} + v^{-2} + 1$$

(Alt) $J(v) = (v^3 - 2v)(v^{-4} + v^{-2}) = -v^{-1} - 2v^{-3} - 2v^{-1} + v$

$$= v - 3v^{-1} - 2v^{-3}$$

(19) If $y = \frac{\sqrt{x}}{2+x}$ differentiating this equation with respect to x

$$\frac{d}{dx}y = \frac{d}{dx}\frac{\sqrt{x}}{2+x} = \frac{(2+x)\frac{1}{2\sqrt{x}} - \sqrt{x}}{(2+x)^2} = \frac{\frac{2+x-2x}{2\sqrt{x}}}{(2+x)^2}$$

$$= \frac{2-x}{2\sqrt{x}(2+x)^2}$$

(25) $y = \frac{\cos x}{1-\sin x} \quad \therefore y' = \frac{(1-\sin x)\sin x - (\cos x)\cdot(1-\cos x)}{(1-\sin x)^2}$

$= \frac{-\sin x + \sin^2 x + \cos^2 x}{(1-\sin x)^2} = \frac{1}{1-\sin x}$

(27) $y = e^x \cosh x = e^x \cdot \frac{e^x + e^{-x}}{2} = \frac{e^{2x}+1}{2}$

$= \frac{\frac{d}{dx} e^{2x} + 0}{2} = \frac{2e^{2x}}{2} = e^{2x}$

Chain Rule

(8) $y = \tanh\sqrt{x}$ let us consider $\sqrt{x} = z \quad \therefore y = \tanh z$

$\therefore \frac{dy}{dz} = \operatorname{sech}^2 z \quad \therefore \frac{dz}{dx} = \frac{1}{2\sqrt{x}}$ &

$\therefore \frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{1}{2\sqrt{x}} = \frac{1}{2\sqrt{x}} \cdot \operatorname{sech}^2 \sqrt{x}$

(6) $y = x \tan^{-1}\sqrt{x}$ let us consider $\sqrt{x} = z \quad \therefore y = x \tan z$

$\therefore \frac{dy}{dz} = \frac{d}{dz} x\tan^{-1} z \quad \therefore \frac{d}{dz}\sqrt{x} = \frac{1}{2\sqrt{x}}$

$= \tan z + \frac{\sec^2 z \cdot 1}{2\sqrt{x}} \quad \frac{\sec^2 z + \tan^{-1} z}{(1+\tan^{-1} z)^2}$ &

$= \frac{x \cdot \frac{d}{dx}\tan^{-1}\sqrt{x} + \tan^{-1}\sqrt{x}\frac{d}{dx}x}{1+(\sqrt{x})^2} = x\frac{1}{(1+\sqrt{x})^2}\frac{1}{2\sqrt{x}} + \tan^{-1}\sqrt{x}$

$= \frac{\sqrt{x}}{2(1+x)} + \tan^{-1}\sqrt{x}$

(17) $y = \sinh(\ln t)$ let us consider $\ln t = z$

$\therefore \frac{d}{dx} y = \cosh(\ln t) \cdot \frac{1}{t} = \frac{e^{\ln t} + e^{-\ln t}}{2} \cdot \frac{1}{t}$

$= \frac{e^{\ln t} + e^{\ln t^{-1}}}{2} \cdot \frac{1}{t} = \frac{t + t^{-1}}{2} \cdot \frac{1}{t} = \frac{t^2+1}{2t^2}$

(16) $y = \ln(\sinh t) \quad \therefore y' = \frac{1}{\sinh t} \cdot \cosh t = \coth t$

$\left(\text{or, } \coth t = \frac{\cosh t}{\sinh t} = \frac{e^t + e^{-t}}{2} \cdot \frac{2}{e^t - e^{-t}} = \frac{e^t + e^{-t}}{e^t - e^{-t}}\right)$

(2) $y = \sqrt{x^x}$ or, $y = (x^x)^{\frac{1}{2}}$ or, $y = x^{\frac{x}{2}}$

taking $\ln$ on both side, $\ln y = \ln x^{\frac{x}{2}}$

or, $\ln y = \frac{x}{2} \ln x \quad \therefore \frac{1}{y}\frac{d}{dx} = \frac{x}{2}\frac{d}{dx}\ln x + \ln x \frac{d}{dx}\frac{x}{2}$

or, $\frac{dy}{dx} = y\left(\frac{1}{2} + \frac{\ln x}{2}\right) = \sqrt{x^x}\left(\frac{1+\ln x}{2}\right)$

(9) $y = (\ln x)^{\cos x} = \cancel{\cos x \ln x}\ \ln(\ln x)^{\cos x} = \cos x \cdot \ln(\ln$

$\therefore \frac{dy}{dx} = \cos x \frac{1}{\ln x} \cdot \frac{1}{x} + \ln(\ln x)(-\sin x)$

$= y\left\{\frac{\cos x}{x \ln x} - \sin x \cdot \ln(\ln x)\right\}$

$= (\ln x)^{\cos x}\left\{\frac{\cos x}{x \ln x} - \sin x \cdot \ln(\ln x)\right\}$

Implicit Differentiation

The derivative of the function of the form $f(x,y)=0$ form $f(x,y)=0$ can be obtained by differentiating each term seperately and solving for $\frac{dy}{dx}$.

$f(x,y) = 0 \qquad x^2y + 3xy^3 - x = 3$

$\left(y\frac{d}{dx}x^2 + x^2\frac{dy}{dx}\right) + \left(3y^3\frac{dx}{dx} + 3x\frac{d\cdot y^3}{dx}\right) - 1 = 0$

or, $2xy + x^2\frac{dy}{dx} + 3y^3 + 3x \cdot 3y^2\frac{dy}{dx} - 1 = 0$

or, $2xy + 3y^3 - 1 + x^2\frac{dy}{dx} + 9xy^2\frac{dy}{dx} = 0$

or, $\frac{d}{dx}(x^2+9y^2) = 1 - 2xy - 3y^3$ or, $\frac{dy}{dx} = \frac{1-2xy-3y^3}{x^2+9xy^2}$

Parametric Differentiation

$x = f(t), \quad y = g(t) \qquad \frac{dy}{dx} = \frac{dy/dt}{dx/dt}$

$x = t^3 - 3t, \quad y = t^3 - 3t \qquad \frac{dy}{dt} = 3t^2 - 6t$

$\frac{dx}{dt} = 3t - 3 \qquad \frac{dy}{dx} = \frac{3t^2-6t}{3t^2-3} = \frac{t^2-2t}{t^2-1}$

$x^2 - 4xy + y^2 = 4$

$2x - 4\left(\frac{d}{dx}\right) \frac{d}{dx} 4xy + \frac{d}{dx} y^2 + \frac{d}{dx} 4 = 0$

or, $2x - \left(4x \cdot \frac{dy}{dx} + 4y \frac{d}{dx} x\right) + 2y \cdot \frac{dy}{dx} = 0$

or, $2x - 4x \cdot \frac{dy}{dx} - 4y + 2y \frac{dy}{dx} = 0$

or, $\frac{dy}{dx}(2y - 4x) = 4y - 2x$ or, $\frac{dy}{dx} = \frac{2y - x}{y - 2x}$

$xe^y = x - y \Rightarrow xe^y - x + y = 0$

$\Rightarrow \frac{d}{dx} xe^y - \frac{d}{dx} x + \frac{d}{dx} y = 0$

$\Rightarrow e^y + x \frac{d}{dx} e^y - 1 + \frac{d}{dx} y = 0$

$\Rightarrow e^y + xe^y \cdot \frac{dy}{dx} - 1 + \frac{dy}{dx} = 0$

$\Rightarrow \frac{dy}{dx}(xe^y + 1) = 1 - e^y \Rightarrow \frac{dy}{dx} = \frac{1 - e^y}{1 + xe^y}$